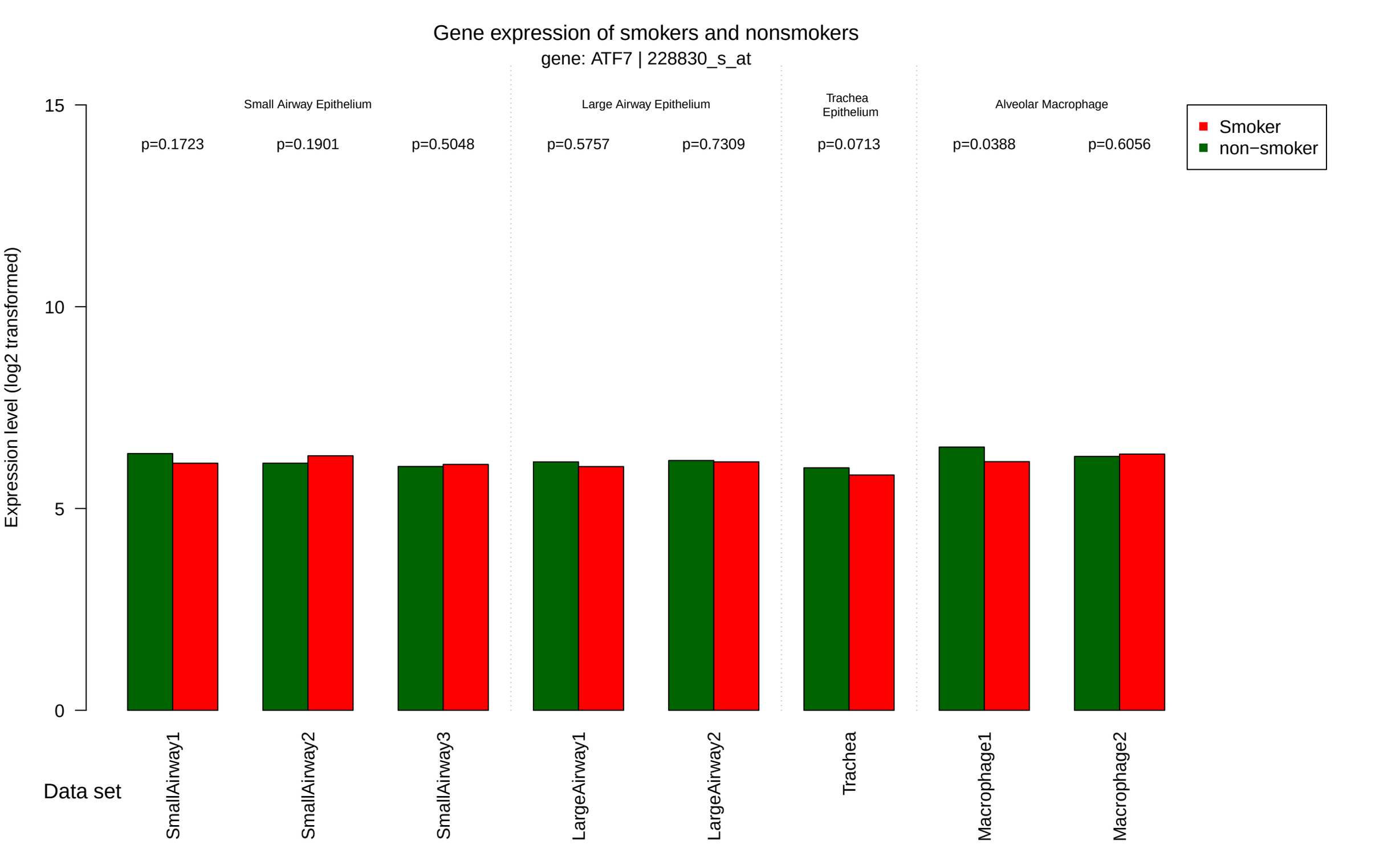
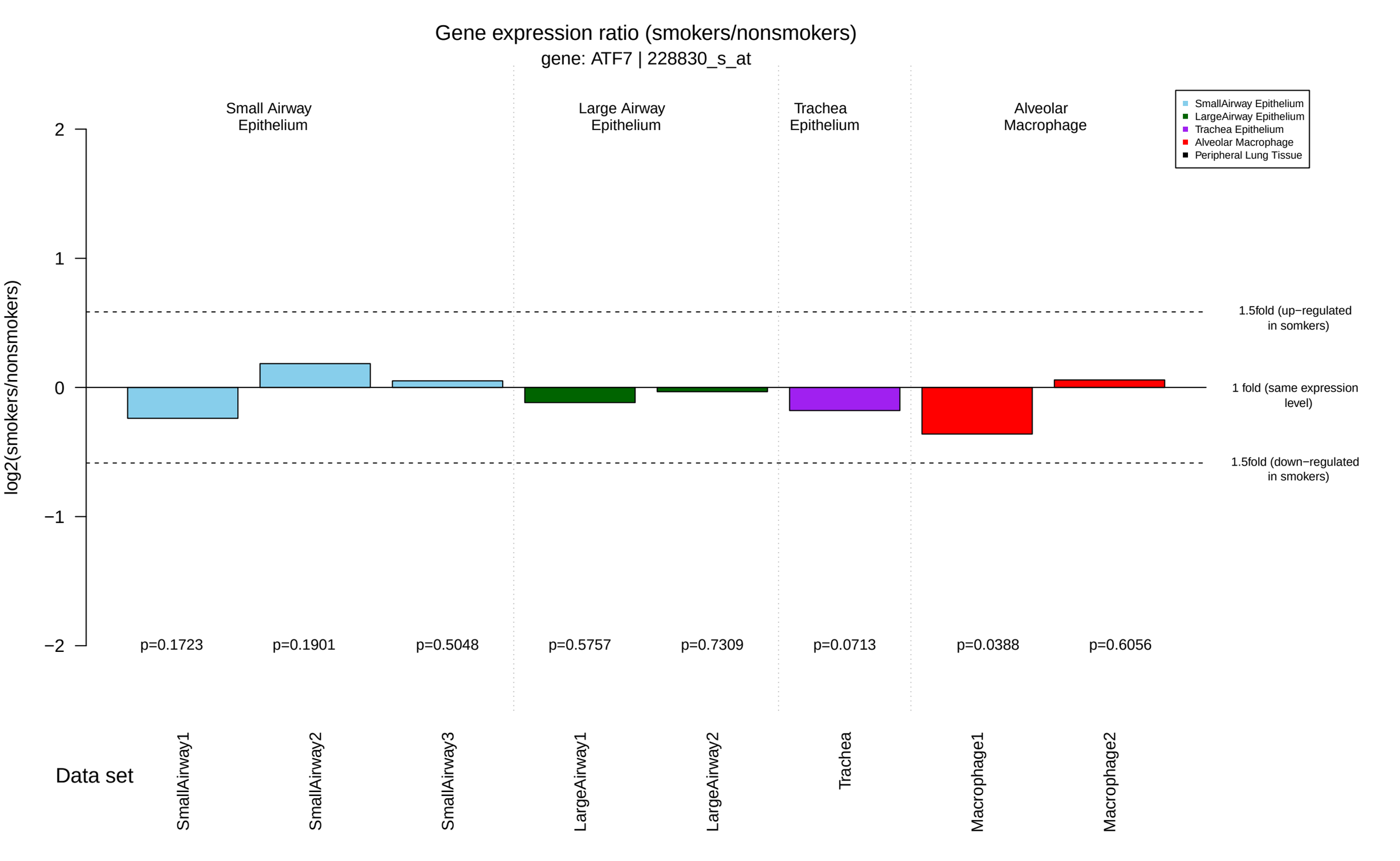
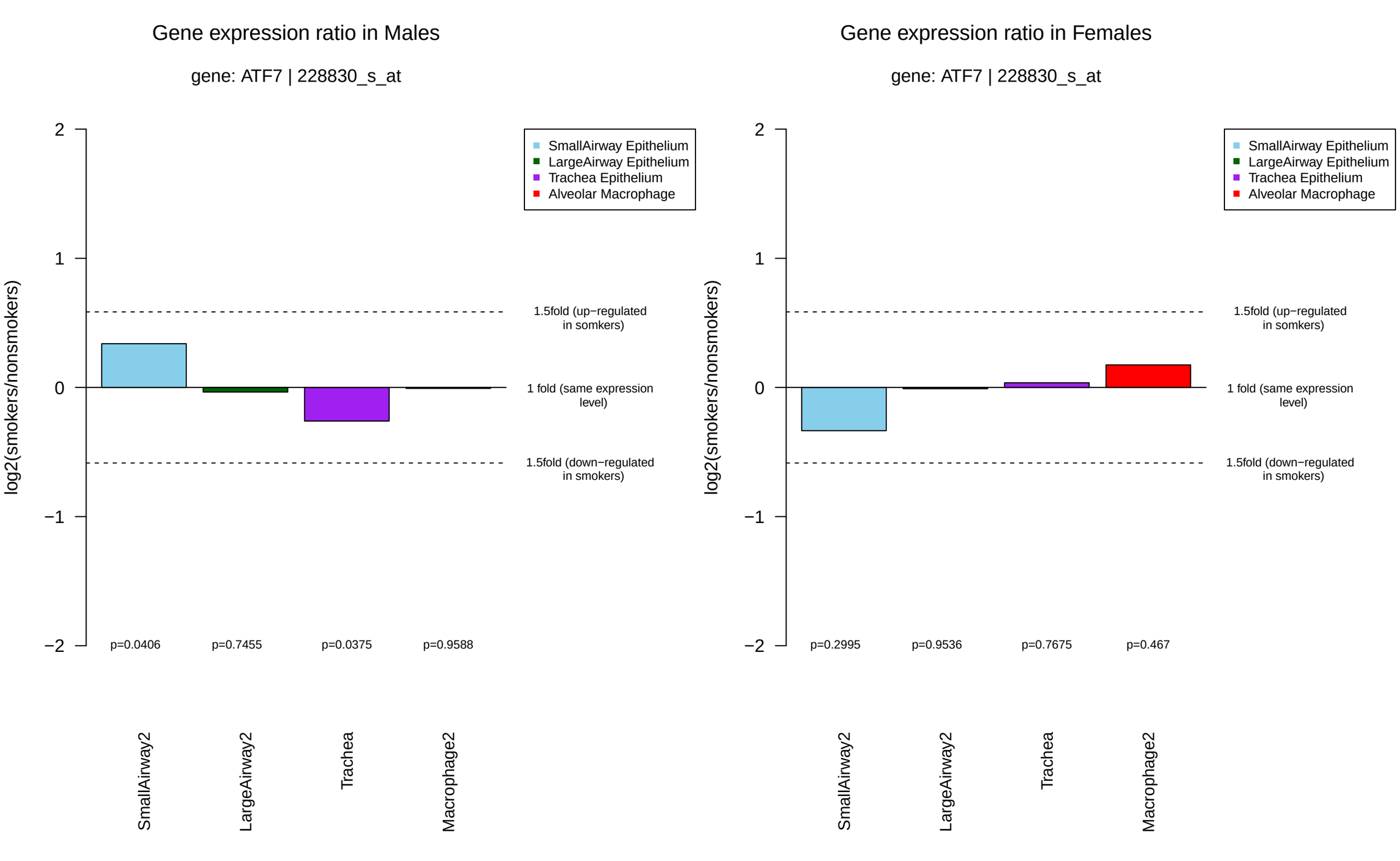
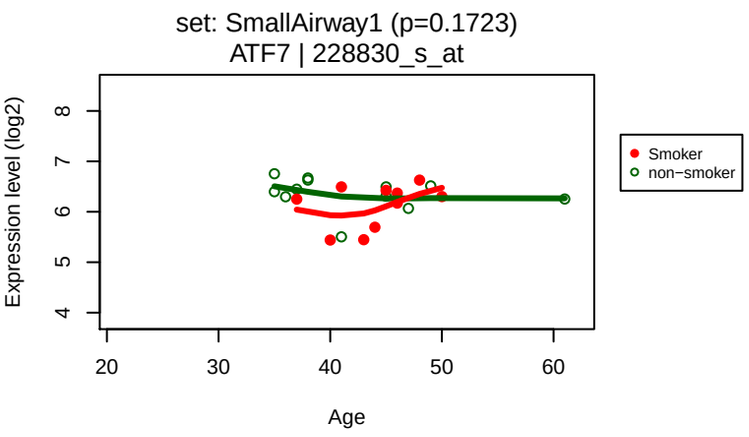
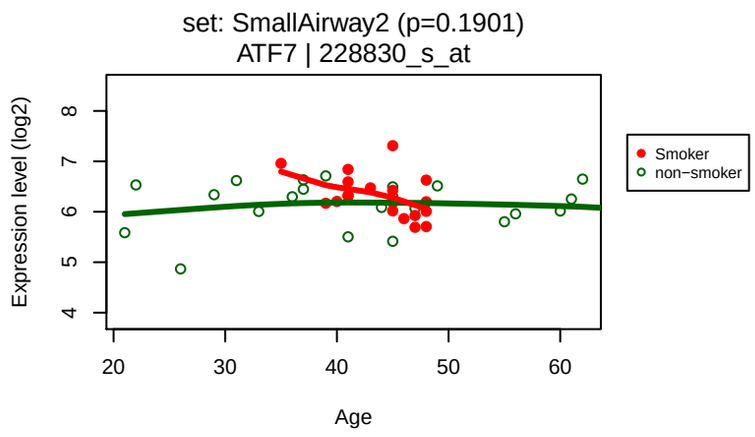
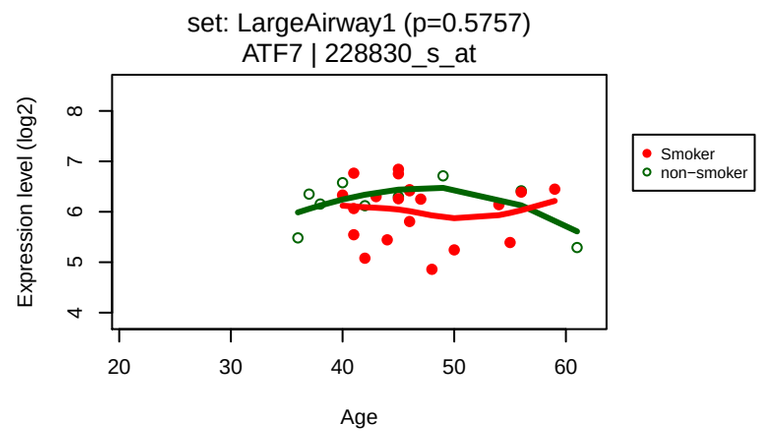

Gene expression of smokers and nonsmokers
gene: ATF7 | 228830_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1723
p=0.1901
p=0.5048
p=0.5757
p=0.7309
p=0.0713
p=0.0388
p=0.6056
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

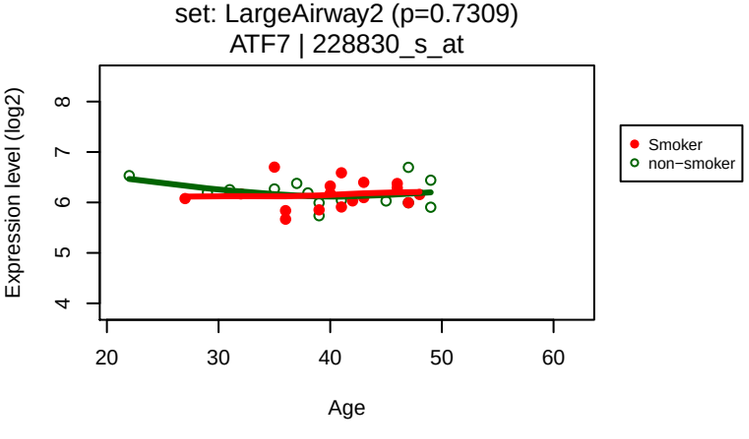

Gene expression ratio (smokers/nonsmokers)
gene: ATF7 | 228830_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.1901
p=0.5048
p=0.5757
p=0.7309
p=0.0713
p=0.0388
p=0.6056
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

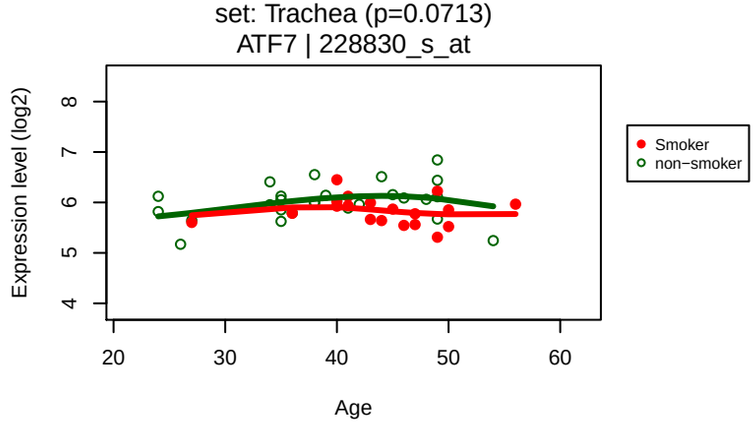

Gene expression ratio in Males
gene: ATF7 | 228830_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0406
p=0.7455
p=0.0375
p=0.9588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF7 | 228830_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2995
p=0.9536
p=0.7675
p=0.467
SmallAirway2
LargeAirway2
Trachea
Macrophage2

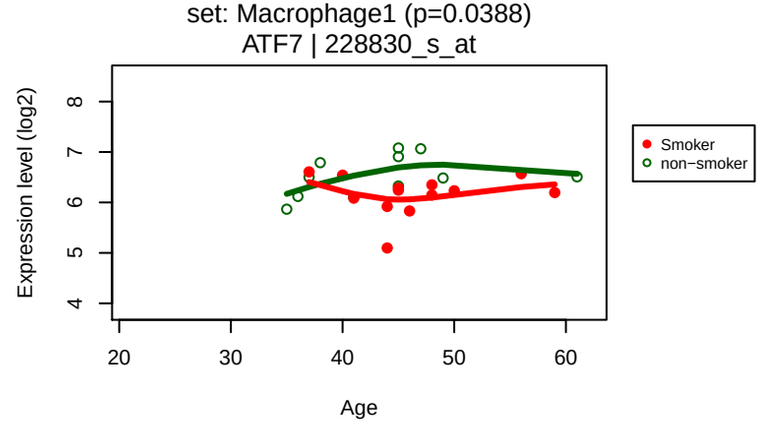

set: SmallAirway1 (p=0.1723)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker

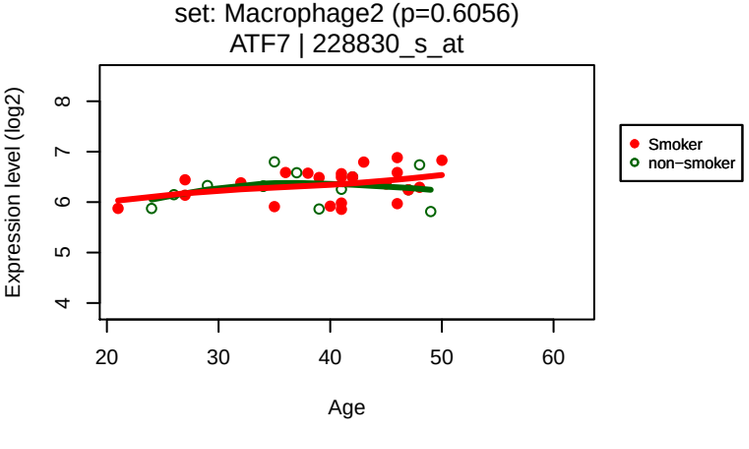

set: SmallAirway2 (p=0.1901)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker

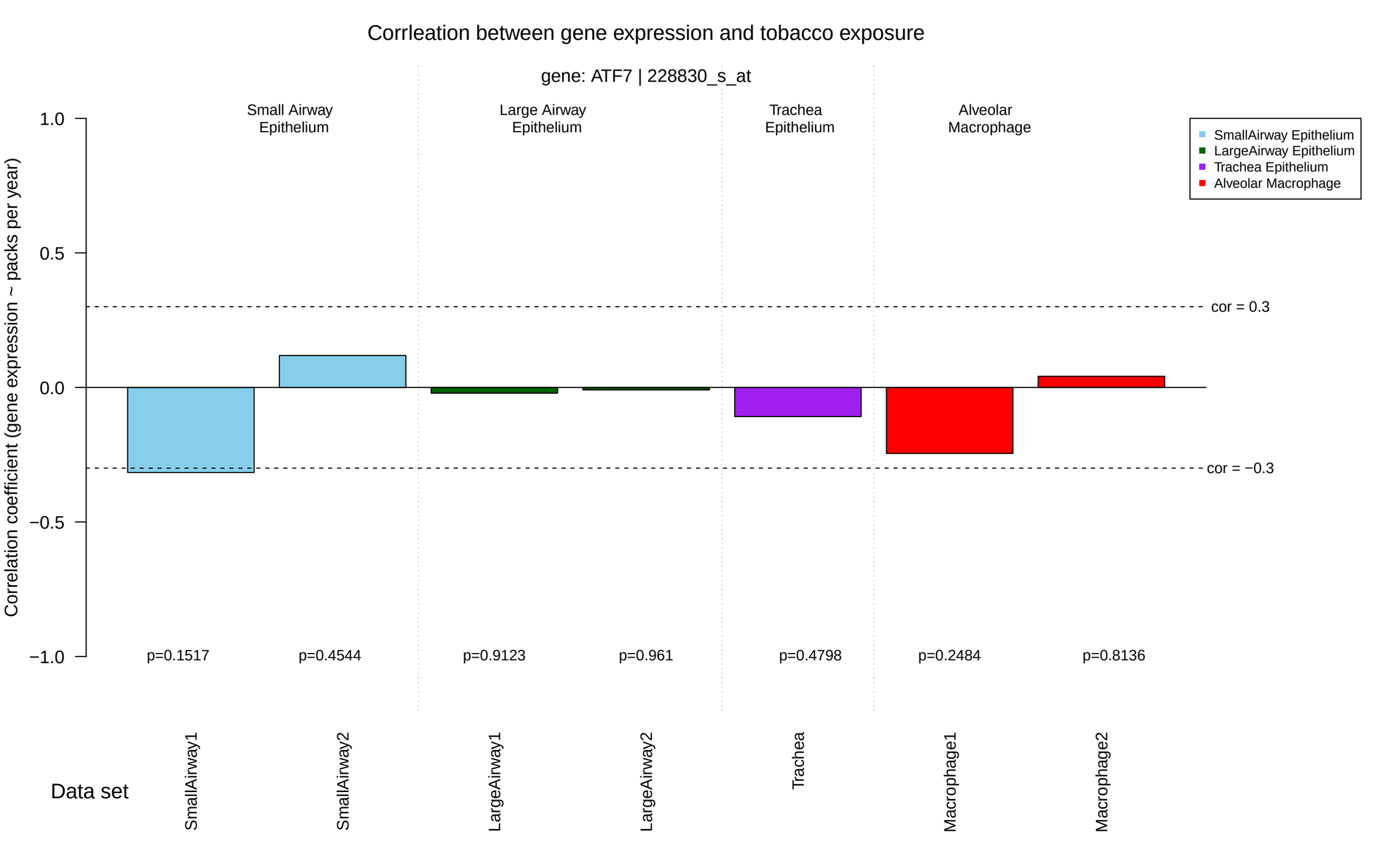

set: LargeAirway1 (p=0.5757)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7309)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0713)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0388)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6056)
ATF7 | 228830_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF7 | 228830_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1517
p=0.4544
p=0.9123
p=0.961
p=0.4798
p=0.2484
p=0.8136
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2